

Statement of Work (SOW)

Drip Controller Automation Software

1. Objective

- Develop an automated Drip Controller system for a common tank with 4 solenoid valves.
- System shall provide Fogger cyclic control, Sprinkler cyclic control, and EC based dosing control.
- System shall support user configurable setpoints through HMI.

2. System Overview

- The system shall operate 4 solenoid valves from a single common tank.
- The system shall support Automatic, Manual, and Stop modes.
- The system shall continuously monitor EC value and control dosing accordingly.

3. HMI Requirements

1. HMI shall display:

- Current EC value
- EC MIN / MAX setpoints
- Fogger, Sprinkler, EC, Valve-4 status
- System mode (Auto/Manual/Stop)
- Active timers

2. A dedicated Setpoints Window shall be provided for configuration of all parameters.

4. Solenoid Valve Configuration

- Valve-1 shall be assigned to Fogger.
- Valve-2 shall be assigned to Sprinkler.
- Valve-3 shall be assigned to EC Control.
- Valve-4 shall remain spare for future use.
- All valves shall operate from a common tank system.

5. Fogger Control

- Fogger shall operate in cyclic mode at full capacity.
- User shall configure ON time, OFF time, and enable/disable.
- System shall execute continuous cycle:

ON → OFF → Repeat

- Example:
 - ON: 30 seconds
 - OFF: 5 minutes

6. Sprinkler Control

- Sprinkler shall operate in cyclic mode at full capacity.
- User shall configure ON time, OFF time, and enable/disable.
- System shall execute continuous cycle:

ON → OFF → Repeat

- Example:
 - ON: 2 minutes
 - OFF: 20 minutes

7. EC Control System

- System shall continuously monitor EC value.
- EC control shall operate based on MIN/MAX logic.
- User shall configure:
 - EC MIN setpoint
 - EC MAX setpoint
 - EC capacity
 - EC ON/OFF cycle time
 - Enable/Disable
- Control logic:
 - If $EC < EC\ MIN \rightarrow EC\ Control\ ON$
 - If $EC \geq EC\ MAX \rightarrow EC\ Control\ OFF$
- EC dosing shall operate in cyclic mode with user defined capacity.
- Example:
 - $EC\ MIN = 1.5\ mS/cm$
 - $EC\ MAX = 2.0\ mS/cm$
 - $ON = 30\ sec$
 - $OFF = 5\ min$

8. Setpoints Configuration

- User shall configure all parameters from Setpoints Window.
- Fogger:
 - ON time
 - OFF time
 - Enable/Disable
- Sprinkler:
 - ON time
 - OFF time
 - Enable/Disable
- EC Control:
 - EC MIN

- EC MAX
 - Capacity
 - Cycle time
 - Enable/Disable
- Valve-4:
 - Reserved for future configuration

9. Operating Modes

- Automatic Mode → Full system automatic operation
- Manual Mode → Individual valve control
- Stop Mode → System OFF state

10. System Operation Logic

- System shall read and validate all setpoints.
- System shall execute Fogger cyclic operation.
- System shall execute Sprinkler cyclic operation.
- System shall continuously monitor EC value.
- System shall activate EC dosing when required.
- System shall stop EC dosing at MAX level.
- System shall repeat all operations continuously.

11. Manual Control

- Manual control shall be available for:
 - Fogger
 - Sprinkler
 - EC Valve
 - Valve-4
- Manual control shall be active only in Manual Mode.

12. Alarms & Warnings

- System shall generate alarms for:
 - EC sensor failure
 - Invalid EC values
 - EC out of range
 - Valve fault
 - Communication failure
 - Tank low level (if sensor available)
- All alarms shall be displayed on HMI.

13. Data Logging (Optional)

- System may log:
 - EC values
 - Valve status
 - Mode status
 - Alarm history
 - Time stamps

14. Tank System

- All valves shall operate from a single common tank.
- Tank level monitoring shall be optional based on sensor availability.

15. Parameter Validation

- System shall validate all user inputs.
- System shall block invalid configurations such as:
 - $EC\ MIN > EC\ MAX$

- Negative timer values
- Invalid capacity values

16. Requirement Summary

- System shall provide:
 - 4 valve automation system
 - Fogger cyclic control
 - Sprinkler cyclic control
 - EC MIN/MAX dosing control
 - User configurable HMI
 - Manual + Automatic modes
 - Alarm system
 - Optional data logging